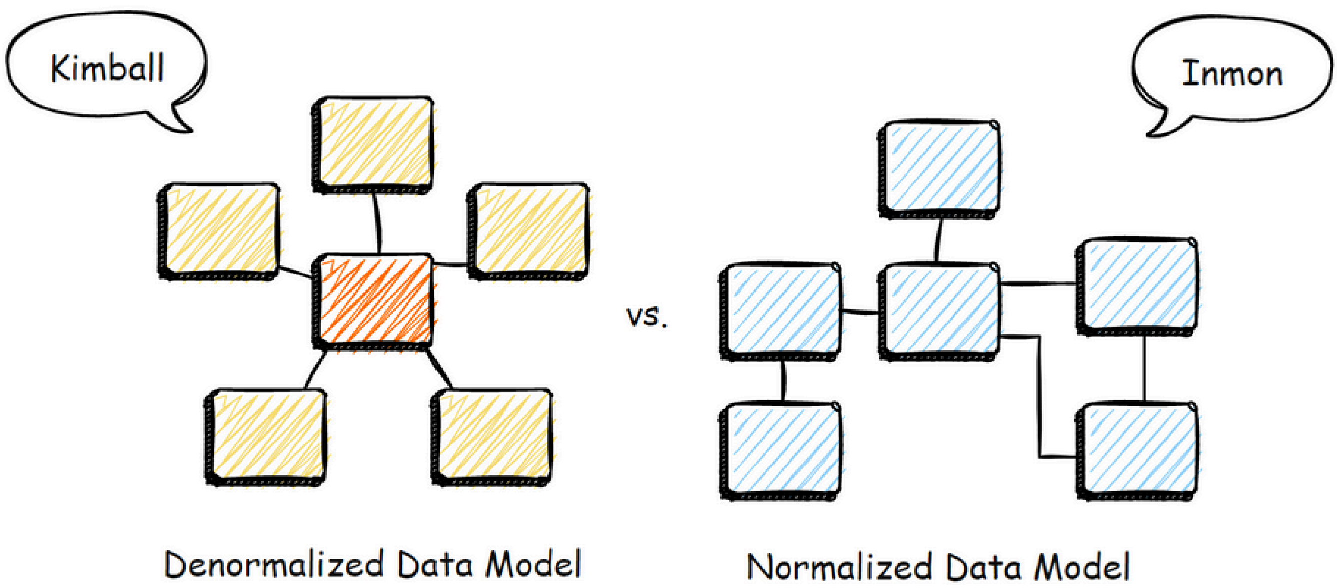
Data Ware House Lab1

Inmon vs Kimball Approaches

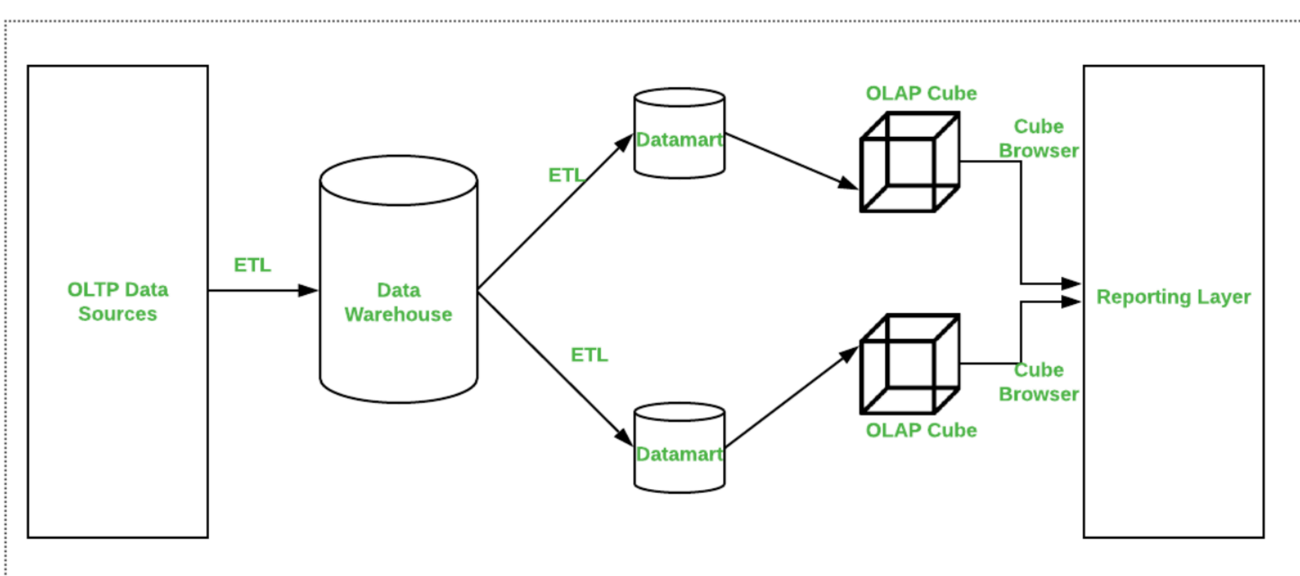


There are two famous methodologies for designing a Data Warehouse:

- **Inmon**: Top-Down approach — starts with a central data warehouse (EDW).
  Like Normalized Data Model.
- **Kimball**: Bottom-Up approach — starts with departmental Data Marts.
  Like Denormalized Data Model

The core difference lies in how data is structured and where the process begins .

Inmon Approach (Top-Down)



Inmon Model

Builds a centralized Enterprise Data Warehouse (EDW) first. Then, Data Marts are created for each business function by extracting data from the EDW.

Advantages

- High data consistency and integration.(جوده البيانات والتكامل بينها عالي)
- Suitable for large and complex organizations.
- Data is normalized (3NF), which reduces redundancy.

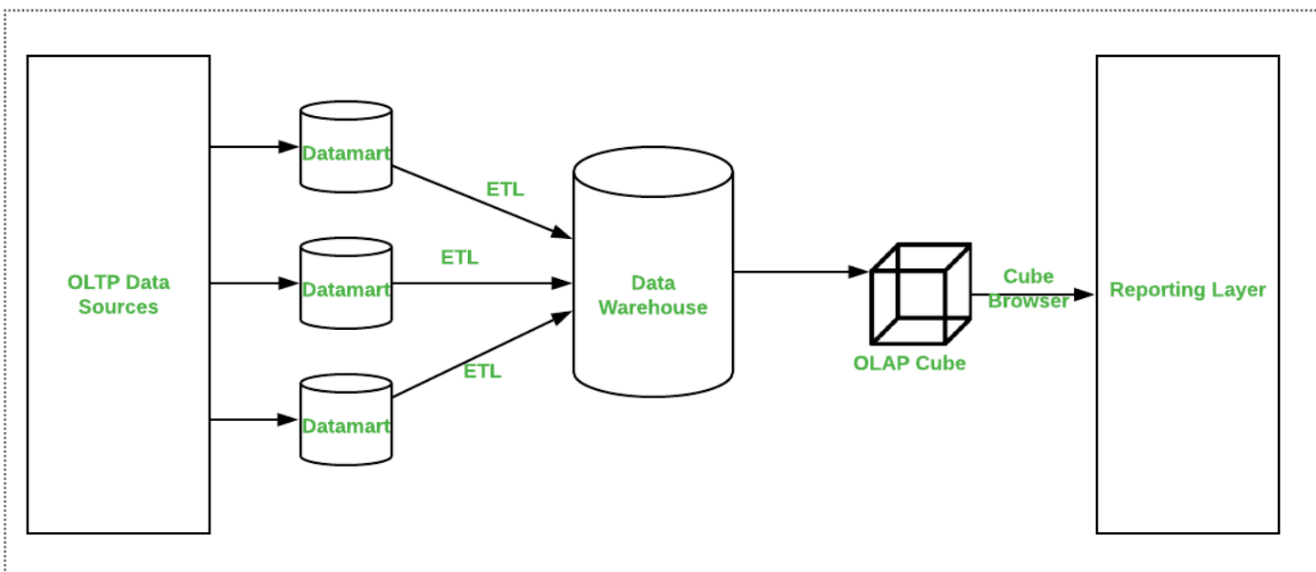
Disadvantages

- Longer implementation time.
- Higher initial cost.
- Less flexible for end-user reporting early on.

When to Use ?

- In large enterprises with complex data sources.
- When long-term data governance is a priority.
- When data consistency is more important than speed.

Kimball Approach (Bottom-Up)



Kimball Model

Begins with the development of Data Marts for each department (e.g., Sales, HR). These are later combined into a complete Data Warehouse.

Advantages

- Faster to implement.
- More user-friendly for reporting and analytics.
- Uses dimensional modeling (star schema), which is easy to understand.

Disadvantages:

- Possible data redundancy.
- Harder to enforce consistency across departments.
- May cause issues when scaling without proper planning.

When to Use ?

- In small to medium-sized companies.

- When quick results and insights are needed.
- When user-friendliness is a priority.

Inmon vs Kimball

Aspect	Inmon (Top-Down)	Kimball (Bottom-Up)
Starting Point	Enterprise Data Warehouse (EDW)	Departmental Data Marts
Modeling	3NF (Normalized)	Dimensional (Star Schema)
Speed of Delivery	Slower	Faster
Data Integration	Strong, centralized	Moderate, decentralized
User-Friendliness	Less in early stages	High from the beginning
Best Fit For	Large, complex organizations	Smaller, agile teams

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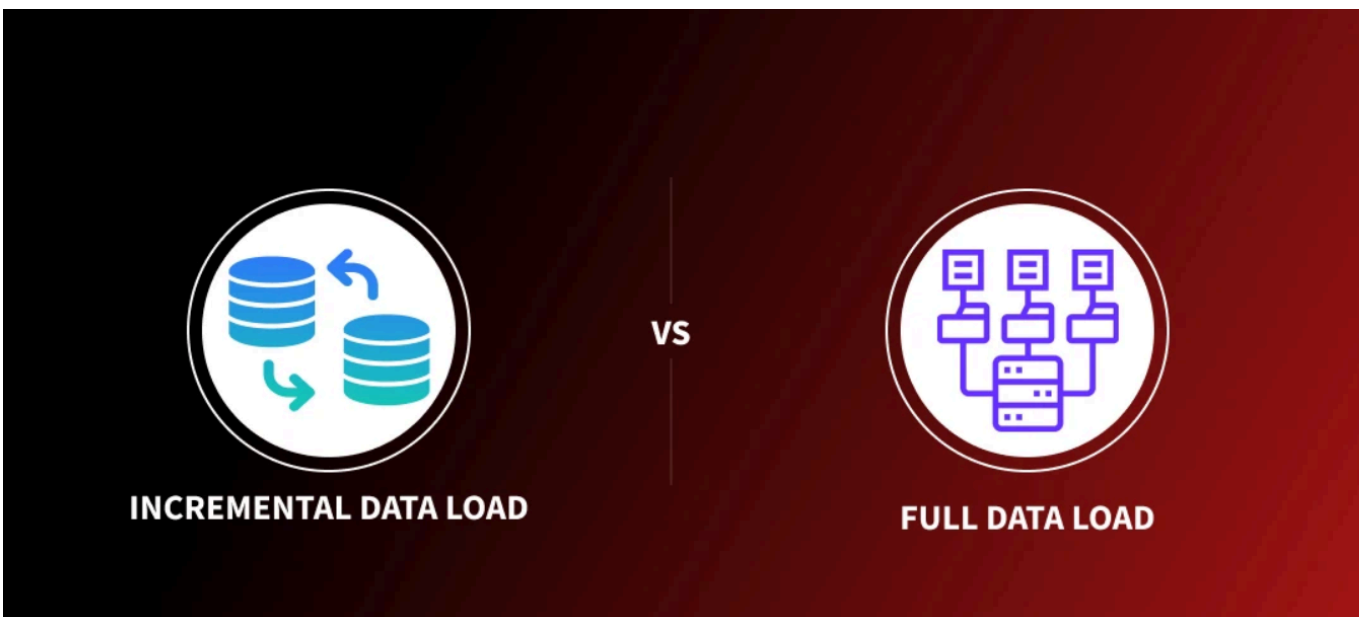
Full Load vs Incremental Load



When transferring data into a Data Warehouse, there are two main loading strategies:

- **Full Load**: Reloads all data from the source each time.
- **Incremental Load**: Loads only new or changed records since the last load.

Full Load

All existing data in the target table is deleted and replaced with the full dataset from the source system.

Advantages

- Simple to design and implement.

- Ensures data consistency without needing change tracking.

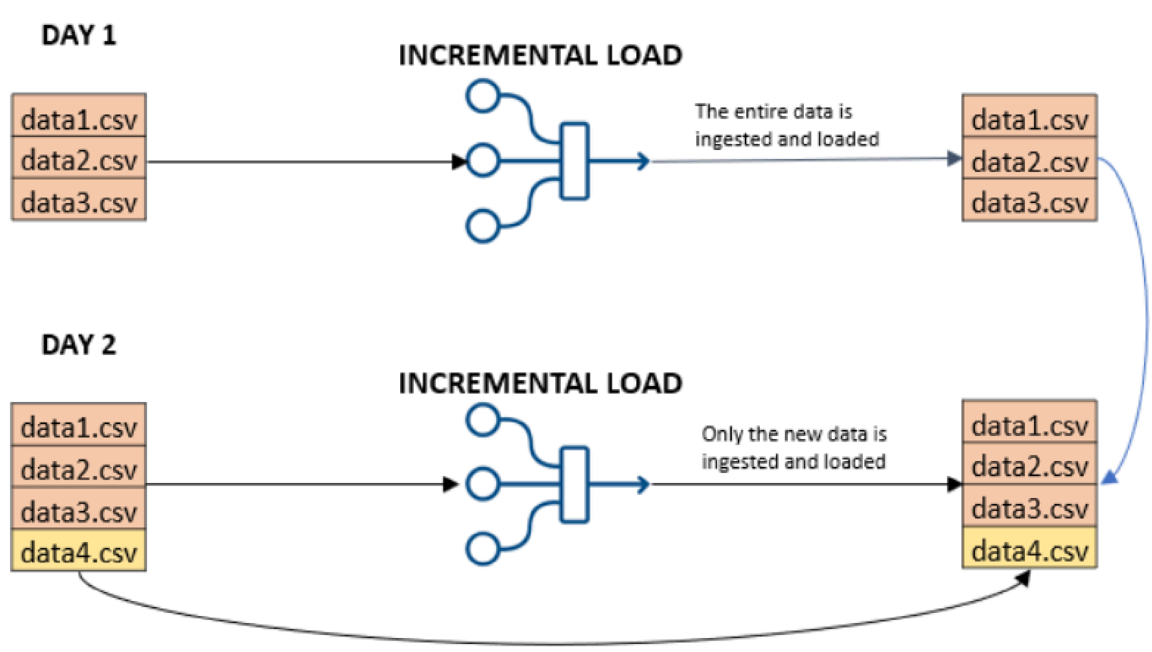
Disadvantages

- Slow and resource-heavy for large datasets.
- Can cause downtime or performance issues.

When to Use

- For small datasets or static data.
- During initial loads or data warehouse testing.
- When change tracking is not available.

Incremental Load



Only loads new or updated data since the last load operation.

Advantages

- Much faster and more efficient.
- Saves system resources and time.
- Ideal for frequent and ongoing data updates.

Disadvantages

- Requires mechanisms to track changes (e.g., timestamps, CDC).
- More complex to implement and maintain.

When to Use ?

- For large datasets.
- In production environments.
- When regular data updates are required.

Full vs Incremental Load

Aspect	Full Load	Incremental Load
Data Volume	All data reloaded	Only new/changed data
Performance	Slower	Faster
Complexity	Simpler	More complex
Ideal Use Case	Small datasets, testing	Large datasets, live systems
Risk of Duplicates	Low	Needs handling to avoid duplicates